## Binary Information

Binary Name => eyecandylocked
Language => C/C++
Arch => x86x64
Platform => Unix/Linux

```
$ file EyeCandyLOCKED
EyeCandyLOCKED: ELF 64-bit LSB executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, for GNU/Linux 2.6.32, BuildID[sha1]=a2f07f96d62e5485980091b97aea74ddbe54648e, with debug_info, not stripped
```

## Analysis

### main() decompiled code

Below is link for the decompiled code of **main()** method by Ghidra.

https://gist.github.com/SHROUDSOURAV/88725fb3323fc8ddfc47555415d42ba4

**Note: There are 2 files in the link main() and checkIt().**

#### *What does this Decompiled Code tells us ???*

- The program uses **PTRACE()** which is a mechanism used for anti debugging so if we try to setup breakpoints and try to reverse engineer the program it will print **"I am a 'crack me' program for God's sake.. Debugging tools like that..?some respect ;)here."** and then the program will exit.
- The program takes a user input string and calls **checkIt()** function where the password checking basically happens.
- If the password we enter is correct then the program prints **"did it broughhh!! :D High five dude! :D.. here is your prize :3"** else **"password!!again fella or type "exit" to.... should I really explain this?"**

### checkIt() decompiled code

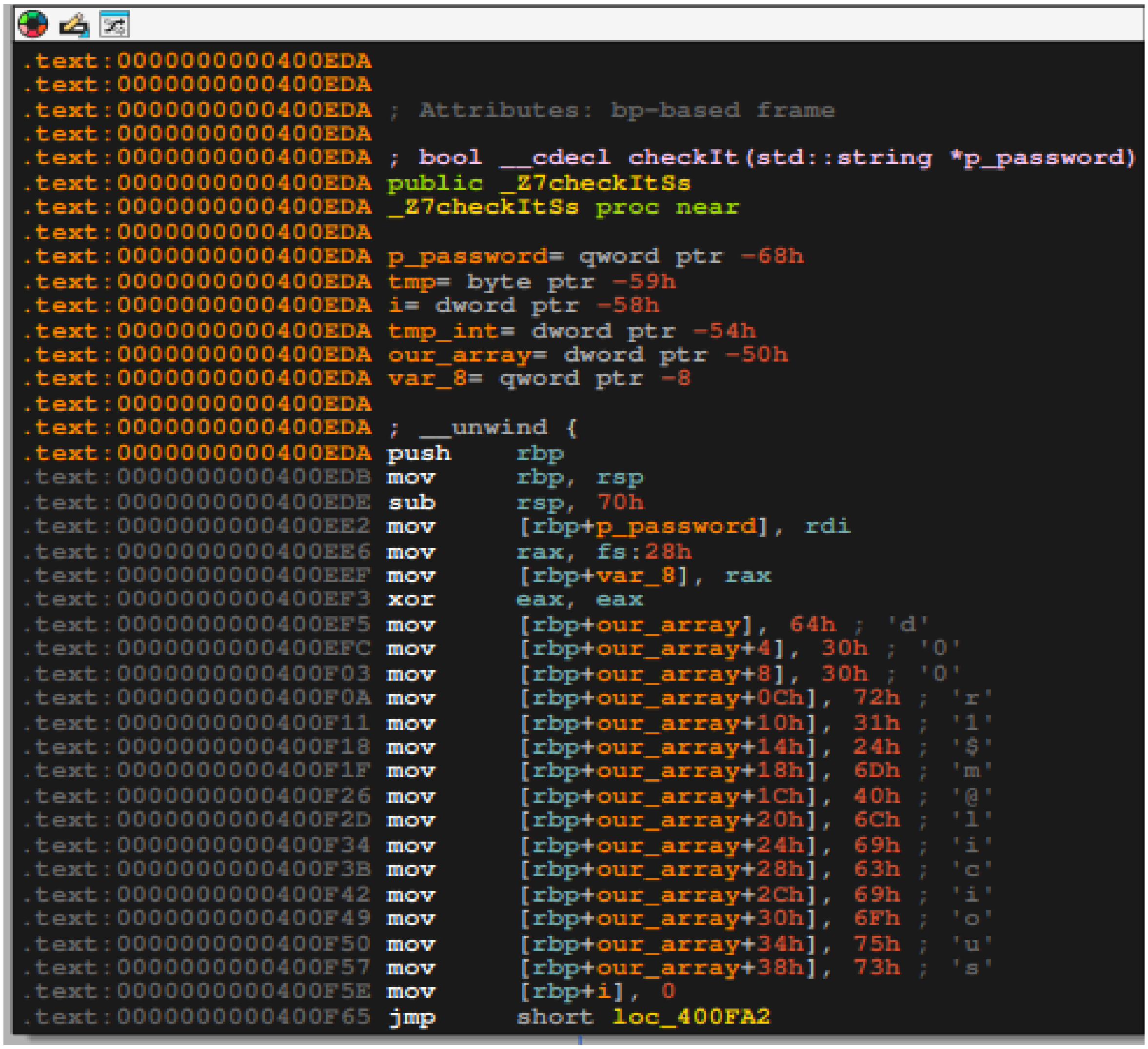
Below is link for the **checkIt()** decompiled code by Ghidra.

https://gist.github.com/SHROUDSOURAV/88725fb3323fc8ddfc47555415d42ba4

**Note: There are 2 files in the link main() and checkIt().**

So we have got the decompiled code for the **checkIt()** function, we can clearly see its comparing each character to those values. Those are hex values but the password we need to input should be in string format. I checked the disassembly of the **checkIt function** in IDA which clearly associated those values with the corresponding characters.

*checkIt() disassembly Image*



```
.text:0000000000400EDA
.text:0000000000400EDA
.text:0000000000400EDA ; Attributes: bp-based frame
.text:0000000000400EDA
.text:0000000000400EDA ; bool __cdecl checkIt(std::string *p_password)
.text:0000000000400EDA public _Z7checkItSs
.text:0000000000400EDA _Z7checkItSs proc near
.text:0000000000400EDA
.text:0000000000400EDA p_password= qword ptr -68h
.text:0000000000400EDA tmp= byte ptr -59h
.text:0000000000400EDA i= dword ptr -58h
.text:0000000000400EDA tmp_int= dword ptr -54h
.text:0000000000400EDA our_array= dword ptr -50h
.text:0000000000400EDA var_8= qword ptr -8
.text:0000000000400EDA
.text:0000000000400EDA ; __unwind {
.text:0000000000400EDA push    rbp
.text:0000000000400EDB mov     rbp, rsp
.text:0000000000400EDE sub     rsp, 70h
.text:0000000000400EE2 mov     [rbp+p_password], rdi
.text:0000000000400EE6 mov     rax, fs:28h
.text:0000000000400EEF mov     [rbp+var_8], rax
.text:0000000000400EF3 xor     eax, eax
.text:0000000000400EF5 mov     [rbp+our_array], 64h ; 'd'
.text:0000000000400EFC mov     [rbp+our_array+4], 30h ; '0'
.text:0000000000400F03 mov     [rbp+our_array+8], 30h ; '0'
.text:0000000000400F0A mov     [rbp+our_array+0Ch], 72h ; 'r'
.text:0000000000400F11 mov     [rbp+our_array+10h], 31h ; '1'
.text:0000000000400F18 mov     [rbp+our_array+14h], 24h ; '$'
.text:0000000000400F1F mov     [rbp+our_array+18h], 6Dh ; 'm'
.text:0000000000400F26 mov     [rbp+our_array+1Ch], 40h ; '@'
.text:0000000000400F2D mov     [rbp+our_array+20h], 6Ch ; 'l'
.text:0000000000400F34 mov     [rbp+our_array+24h], 69h ; 'i'
.text:0000000000400F3B mov     [rbp+our_array+28h], 63h ; 'c'
.text:0000000000400F42 mov     [rbp+our_array+2Ch], 69h ; 'i'
.text:0000000000400F49 mov     [rbp+our_array+30h], 6Fh ; 'o'
.text:0000000000400F50 mov     [rbp+our_array+34h], 75h ; 'u'
.text:0000000000400F57 mov     [rbp+our_array+38h], 73h ; 's'
.text:0000000000400F5E mov     [rbp+i], 0
.text:0000000000400F65 jmp     short loc_400FA2
```

So, the password for unlocking the door is **d00r1$m@licious**.

## Testing our input

```
$ ./EyeCandyLOCKED

                *** THE DOOR IS LOCKED! ***


Gimme your password: d00r1$m@licious


                ****************************
                *** THE DOOR IS UNLOCKED! ***
                ****************************


You did it broughhh!! :D High five dude! :D
Anyway.. here is your prize :3

      http://cdn.business2community.com/wp-content/uploads/2014/10/Witch-Hat-Cat-
Costume-For-Halloween.jpg.jpg
```